Supplementary Information

Mouth-rinses and SARS-CoV-2 viral load in saliva: A living systematic review

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Supplementary Checklist 1. The PRISMA Statement[1]

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7

Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	NA

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7-8
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	8-9
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	8-9
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	9
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	9
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	11
Conclusions	26	Provide a general interpretation of the results in the	11

		context of other evidence, and implications for future research.	
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	1

[1] Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

Supplementary Table 1. Search Strategies.

Database	**PubMed** **Date:** May 24, 2021	
Search Strategy	#1	SARS-CoV-2[Mesh]
	#2	COVID-19[Mesh]
	#3	Corona Virus[tiab]
	#4	COVID-19[tiab]
	#5	COVID19*[tiab]
	#6	2019-nCoV[tiab]
	#7	SARS-CoV-2[tiab]
	#8	SARS-CoV2[tiab]
	#9	(Pneumonia[tiab] AND Wuhan[tiab] AND 2019[tiab])
	#10	Coronavir*[tiab]
	#11	Coronovir*[tiab]
	#12	Virus Corona[tiab]
	#13	Corono Virus[tiab]
	#14	HCov*[tiab]
	#15	CV19*[tiab]
	#16	CV-19[tiab]
	#17	N-Cov[tiab]
	#18	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17
	#19	Mouthwashes[Mesh]

	#20	Mouth Rinse*[tiab]
	#21	Mouthrinse*[tiab]
	#22	Mouth Bath*[tiab]
	#23	Mouthbath*[tiab]
	#24	Mouth Wash*[tiab]
	#25	Mouthwash*[tiab]
	#26	Oral Rinse*[tiab]
	#27	Oralrinse*[tiab]
	#28	Oral Bath*[tiab]
	#29	Oral Wash*[tiab]
	#30	Oralwash*[tiab]
	#31	Dental Rinse*[tiab]
	#32	Dentalrinse*[tiab] - Schema: all
	#33	Dentalrinse*[tiab]
	#34	Dental Bath*[tiab]
	#35	Dental Wash*[tiab]
	#36	#19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34 OR #35
	#37	#18 AND #36

Database	**CINAHL** **Date:** May 24, 2021	
Search Strategy	S1	(MH "COVID-19")
	S2	(MH "Severe Acute Respiratory Syndrome")
	S3	TI (Coron* N1 Virus) OR AB (Coron* N1 Virus)
	S4	TI COVID-19 OR AB COVID-19
	S5	TI COVID19* OR AB COVID19*
	S6	TI 2019-nCoV OR AB 2019-nCoV
	S7	TI SARS-CoV-2 OR AB SARS-CoV-2
	S8	TI SARS-CoV2 OR AB SARS-CoV2

	S9	TI SARSCoV2 OR AB SARSCoV2
	S10	TI (Pneumonia AND Wuhan AND 2019) OR AB (Pneumonia AND Wuhan AND 2019)
	S11	TI Coronavir* OR AB Coronavir*
	S12	TI HCov* OR AB HCov*
	S13	TI CV19* OR AB CV19*
	S14	TI CV-19* OR AB CV-19*
	S15	TI N-Cov OR AB N-Cov
	S16	S1 OR S2 OR S3 OR S4 OR S5 OR S6 OR S7 OR S8 OR S9 OR S10 OR S11 OR S12 OR S13 OR S14 OR S15
	S17	(MH "Mouthwashes+")
	S18	TI (Mouth N1 Rinse*) OR AB (Mouth N1 Rinse*)
	S19	TI Mouthrinse* OR AB Mouthrinse*
	S20	TI (Mouth N1 Bath*) OR AB (Mouth N1 Bath*)
	S21	TI Mouthbath* OR AB Mouthbath*
	S22	TI (Mouth N1 Wash*) OR AB (Mouth N1 Wash*)
	S23	TI Mouthrinse* OR AB Mouthrinse*
	S24	TI (Mouth N1 Bath*) OR AB (Mouth N1 Bath*)
	S25	TI Mouthbath* OR AB Mouthbath*
	S26	TI (Mouth N1 Wash*) OR AB (Mouth N1 Wash*)
	S27	TI Mouthwash* OR AB Mouthwash*
	S28	TI (Oral N1 Rinse*) OR AB (Oral N1 Rinse*)
	S29	TI Oralrinse* OR AB Oralrinse*
	S30	TI (Oral N1 Bath*) OR AB (Oral N1 Bath*)
	S31	TI (Oral N1 Wash*) OR AB (Oral N1 Wash*)
	S32	TI Oralwash OR AB Oralwash*
	S33	TI (Dental N1 Rinse*) OR AB (Dental N1 Rinse*)
	S34	TI Dentalrinse* OR AB Dentalrinse*
	S35	TI (Dental N1 Bath*) OR AB (Dental N1 Bath*)
	S36	TI (Dental N1 Wash*) OR AB (Dental N1 Wash*)

	S37	S17 OR S18 OR S19 OR S20 OR S21 OR S22 OR S23 OR S24 OR S25 OR S26 OR S27 OR S28 OR S29 OR S30 OR S31 OR S32 OR S33 OR S34 OR S35 OR S36
	S38	S16 AND S37

Database	**Cochrane Library** **Date:** May 24, 2021	
Search Strategy	#1	MeSH descriptor: [SARS-CoV-2] explode all trees
	#2	MeSH descriptor: [COVID-19] explode all trees
	#3	(Coron* NEAR/1 Virus):ti,ab,kw
	#4	COVID-19:ti,ab,kw
	#5	COVID19:ti,ab,kw
	#6	"2019-nCoV":ti,ab,kw
	#7	"SARS-CoV-2":ti,ab,kw
	#8	"SARS-CoV2":ti,ab,kw
	#9	SARSCoV2:ti,ab,kw
	#10	(Pneumonia AND Wuhan AND 2019):ti,ab,kw
	#11	Coronavir*:ti,ab,kw
	#12	Coronovir*:ti,ab,kw
	#13	HCov*:ti,ab,kw
	#14	CV19*:ti,ab,kw
	#15	CV-19*:ti,ab,kw
	#16	"N-Cov":ti,ab,kw
	#17	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16
	#18	MeSH descriptor: [Mouthwashes] explode all trees
	#19	(Mouth NEAR/1 Rinse*):ti,ab,kw
	#20	Mouthrinse*:ti,ab,kw
	#21	(Mouth NEAR/1 Bath*):ti,ab,kw
	#22	Mouthbath*:ti,ab,kw
	#23	(Mouth NEAR/1 Wash*):ti,ab,kw

	#24	Mouthwash*:ti,ab,kw
	#25	(Oral NEAR/1 Rinse*):ti,ab,kw
	#26	Oralrinse*:ti,ab,kw
	#27	(Oral NEAR/1 Bath*):ti,ab,kw
	#28	(Oral NEAR/1 Wash*):ti,ab,kw
	#29	Oralwash*:ti,ab,kw
	#30	(Dental NEAR/1 Rinse*):ti,ab,kw
	#31	Dentalrinse*:ti,ab,kw
	#32	(Dental NEAR/1 Bath*):ti,ab,kw
	#33	(Dental NEAR/1 Wash*):ti,ab,kw
	#34	#18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33
	#35	#17 AND #34

Database	**Embase** **Date:** May 24, 2021	
Search Strategy	#1	'severe acute respiratory syndrome coronavirus 2'/exp
	#2	'coronavirus disease 2019'/exp
	#3	(coron* NEAR/1 virus):ti,ab
	#4	'covid 19':ti,ab
	#5	covid19*:ti,ab
	#6	'2019 ncov':ti,ab
	#7	'sars-cov-2':ti,ab
	#8	'sars-cov2':ti,ab
	#9	'sarscov2':ti,ab
	#10	pneumonia:ti,ab AND wuhan:ti,ab AND 2019:ti,ab
	#11	coronavir*:ti,ab
	#12	coronovir*:ti,ab
	#13	hcov*:ti,ab
	#14	cv19*:ti,ab

	#15	'cv 19*':ti,ab
	#16	'n cov':ti,ab
	#17	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16
	#18	'mouthwash'/exp
	#19	(mouth NEAR/1 rinse*):ti,ab
	#20	mouthrinse*:ti,ab
	#21	(mouth NEAR/1 bath*):ti,ab
	#22	mouthbath*:ti,ab
	#23	(mouth NEAR/1 wash*):ti,ab
	#24	mouthwash*:ti,ab
	#25	(oral NEAR/1 rinse*):ti,ab
	#26	oralrinse*:ti,ab
	#27	(oral NEAR/1 bath*):ti,ab
	#28	(oral NEAR/1 wash*):ti,ab
	#29	oralwash*:ti,ab
	#30	(dental NEAR/1 rinse*):ti,ab
	#31	dentalrinse*:ti,ab
	#32	(dental NEAR/1 bath*):ti,ab
	#33	(dental NEAR/1 wash*):ti,ab
	#34	#18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33
	#35	#17 AND #34

Database	**Scopus** **Date:** May 24, 2021	
Search Strategy	#1	TITLE-ABS-KEY ("SARS-CoV-2")
	#2	TITLE-ABS-KEY ("COVID-19")
	#3	TITLE-ABS-KEY ("Corona Virus")
	#4	TITLE-ABS-KEY (COVID19*)
	#5	TITLE-ABS-KEY ("2019-nCoV")

	#6	TITLE-ABS-KEY ("SARS-CoV2")
	#7	TITLE-ABS-KEY (Pneumonia AND Wuhan AND 2019)
	#8	TITLE-ABS-KEY (Coronavir*)
	#9	TITLE-ABS-KEY (Coronovir*)
	#10	TITLE-ABS-KEY ("Virus Corona")
	#11	TITLE-ABS-KEY ("Corono Virus")
	#12	TITLE-ABS-KEY (HCov*)
	#13	TITLE-ABS-KEY (CV19*)
	#14	TITLE-ABS-KEY ("CV-19")
	#15	TITLE-ABS-KEY ("N-Cov")
	#16	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15
	#17	TITLE-ABS-KEY ("Mouthwashes")
	#18	TITLE-ABS-KEY (Mouth Rinse*)
	#19	TITLE-ABS-KEY (Mouthrinse*)
	#20	TITLE-ABS-KEY (Mouth Bath*)
	#21	TITLE-ABS-KEY (Mouthbath*)
	#22	TITLE-ABS-KEY (Mouth Wash*)
	#23	TITLE-ABS-KEY (Mouthwash*)
	#24	TITLE-ABS-KEY (Oral Rinse*)
	#25	TITLE-ABS-KEY (Oralrinse*)
	#26	TITLE-ABS-KEY (Oral Bath*)
	#27	TITLE-ABS-KEY (Oral Wash*)
	#28	TITLE-ABS-KEY (Oralwash*)
	#29	TITLE-ABS-KEY (Dental Rinse*)
	#30	TITLE-ABS-KEY (Dentalrinse*)
	#31	TITLE-ABS-KEY (Dental Bath*)
	#32	TITLE-ABS-KEY (Dental Wash*)
	#33	#17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32
	#34	#16 AND #33

Database	**Dentistry & Oral Sciences Source** **Date:** May 24, 2021	
Search Strategy	S1	(MH "COVID-19")
	S2	(MH "Severe Acute Respiratory Syndrome")
	S3	TI (Coron* N1 Virus) OR AB (Coron* N1 Virus)
	S4	TI COVID-19 OR AB COVID-19
	S5	TI COVID19* OR AB COVID19*
	S6	TI 2019-nCoV OR AB 2019-nCoV
	S7	TI SARS-CoV-2 OR AB SARS-CoV-2
	S8	TI SARS-CoV2 OR AB SARS-CoV2
	S9	TI SARSCoV2 OR AB SARSCoV2
	S10	TI (Pneumonia AND Wuhan AND 2019) OR AB (Pneumonia AND Wuhan AND 2019)
	S11	TI Coronavir* OR AB Coronavir*
	S12	TI HCov* OR AB HCov*
	S13	TI CV19* OR AB CV19*
	S14	TI CV-19* OR AB CV-19*
	S15	TI N-Cov OR AB N-Cov
	S16	S1 OR S2 OR S3 OR S4 OR S5 OR S6 OR S7 OR S8 OR S9 OR S10 OR S11 OR S12 OR S13 OR S14 OR S15
	S17	(MH "Mouthwashes+")
	S18	TI (Mouth N1 Rinse*) OR AB (Mouth N1 Rinse*)
	S19	TI Mouthrinse* OR AB Mouthrinse*
	S20	TI (Mouth N1 Bath*) OR AB (Mouth N1 Bath*)
	S21	TI Mouthbath* OR AB Mouthbath*
	S22	TI (Mouth N1 Wash*) OR AB (Mouth N1 Wash*)
	S23	TI Mouthrinse* OR AB Mouthrinse*
	S24	TI (Mouth N1 Bath*) OR AB (Mouth N1 Bath*)
	S25	TI Mouthbath* OR AB Mouthbath*
	S26	TI (Mouth N1 Wash*) OR AB (Mouth N1 Wash*)

	S27	TI Mouthwash* OR AB Mouthwash*
	S28	TI (Oral N1 Rinse*) OR AB (Oral N1 Rinse*)
	S29	TI Oralrinse* OR AB Oralrinse*
	S30	TI (Oral N1 Bath*) OR AB (Oral N1 Bath*)
	S31	TI (Oral N1 Wash*) OR AB (Oral N1 Wash*)
	S32	TI Oralwash OR AB Oralwash*
	S33	TI (Dental N1 Rinse*) OR AB (Dental N1 Rinse*)
	S34	TI Dentalrinse* OR AB Dentalrinse*
	S35	TI (Dental N1 Bath*) OR AB (Dental N1 Bath*)
	S36	TI (Dental N1 Wash*) OR AB (Dental N1 Wash*)
	S37	S17 OR S18 OR S19 OR S20 OR S21 OR S22 OR S23 OR S24 OR S25 OR S26 OR S27 OR S28 OR S29 OR S30 OR S31 OR S32 OR S33 OR S34 OR S35 OR S36
	S38	S16 AND S37

Database	**LILACS** **Date:** May 24, 2021	
Search Strategy	#1	(MH SARS-CoV-2 OR MH COVID-19 OR Corona OR COVID-19 OR COVID19$ OR 2019-nCoV OR SARS-CoV-2 OR SARS-CoV2 OR Coronavir$ OR Coronovir$ OR HCov$ OR CV19$ OR CV-19 OR N-Cov) AND (MH Mouthwashes OR Mouthwas$ OR Mouthrinse$ OR Mouthbath$ OR Rinse$ OR Enjuague$ OR Enxague) [Words]

Database	**medRxiv AND bioRxiv** **Date:** May 24, 2021	
Search Strategy	#1	"Mouthwashes AND Covid"

Supplementary References 1. List of included studies in this living systematic review.

Gottsauner MJ, Michaelides I, Schmidt B, Scholz KJ, Buchalla W, Widbiller M, Hitzenbichler F, Ettl T, Reichert TE, Bohr C, Vielsmeier V, Cieplik F. A prospective clinical pilot study on the effects of a hydrogen peroxide mouthrinse on the intraoral viral load of SARS-CoV-2. Clin Oral Investig. 2020;24(10):3707-3713. doi: 10.1007/s00784-020-03549-1.

Yoon JG, Yoon J, Song JY, Yoon SY, Lim CS, Seong H, Noh JY, Cheong HJ, Kim WJ. Clinical Significance of a High SARS-CoV-2 Viral Load in the Saliva. J Korean Med Sci. 2020;35(20):e195. doi: 10.3346/jkms.2020.35.e195.

Martínez Lamas L, Diz Dios P, Pérez Rodríguez MT, Del Campo Pérez V, Cabrera Alvargonzalez JJ, López Domínguez AM, Fernandez Feijoo J, Diniz Freitas M, Limeres Posse J. Is povidone iodine mouthwash effective against SARS-CoV-2? First in vivo tests. Oral Dis. 2020:10.1111/odi.13526. doi: 10.1111/odi.13526.

Seneviratne CJ, Balan P, Ko KKK, Udawatte NS, Lai D, Ng DHL, Venkatachalam I, Lim KS, Ling ML, Oon L, Goh BT, Sim XYJ. Efficacy of commercial mouth-rinses on SARS-CoV-2 viral load in saliva: randomized control trial in Singapore. Infection. 2021;49(2):305-311. doi: 10.1007/s15010-020-01563-9.

Schürmann M, Aljubeh M, Tiemann C, Sudhoff H. Mouthrinses against SARS-CoV-2: anti-inflammatory effectivity and a clinical pilot study. Eur Arch Otorhinolaryngol. 2021:1–9. doi: 10.1007/s00405-021-06873-8.